Group 4 assignment 3

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** Let h be a binary classifier, S ∈ [0, 1] a score, Y ∈ {0, 1} the outcome, A ∈ {0, 1} a protected attribute**

1.State the conditions for Calibration, FPR parity, and FNR parity.

Calibration

Interpretation

Calibration (predictive parity) requires that the predicted risk score has the same meaning across different groups, which can be written as: $P(Y = 1 | S = s, A = 0) = P(Y = 1 | S = s, A = 1)$,where Y is the true outcome, S is the predicted score, and A is a protected attribute such as race.

Intuitively, this means that for the same score s, the actual probability of the outcome should be the same regardless of group membership. For example, if a person is assigned a 70% risk score, then:

- White individuals with this score, about 70% should truly experience the outcome
- Black individuals with the same score, about 70% should also experience the outcome

In this sense, the score is equally interpretable across groups.

Real-word Application

This type of fairness is commonly used in real-world risk prediction settings such as criminal justice (e.g., COMPAS predicting recidivism), credit scoring (predicting default risk), healthcare risk models, and insurance pricing, where the reliability of predicted probabilities is crucial.

Limitation

However, even if a model is well-calibrated, the outcomes can still be unfair. This is because calibration only ensures that the scores reflect true probabilities, but does not guarantee equal error rates across groups.

For instance, if the base rates differ between groups (e.g., one group has a higher overall recidivism rate than another), a calibrated model may still produce higher false positive rates for one group and higher false negative rates for another.

Equalized Odds FPR & FNR

Interpretation

FPR parity and FNR parity require that error rates are equal across groups, which is often referred to as equalized odds. This can be written as: $TPR_0 = TPR_1 \; and \; FNR_0 = FNR_1$

Intuitively, this means that the model treats mistakes "equally" across groups.

- **FPR (False Positive Rate)**: among people who will not experience the outcome, the probability of being incorrectly predicted as positive should be the same across groups
- **FNR (False Negative Rate)**: among people who will experience the outcome, the probability of being incorrectly predicted as negative should also be the same across groups

For example:

- If 20% of White individuals who will not reoffend are incorrectly labeled as high risk, then about 20% of Black individuals who will not reoffend should also be labeled as high risk (equal FPR)
- If 10% of White individuals who will reoffend are incorrectly labeled as low risk, then about 10% of Black individuals who will reoffend should also be labeled as low risk (equal FNR)

Real-word Application

This type of fairness is commonly emphasized in high-stakes settings such as criminal justice (e.g., COMPAS), where unequal error rates may lead to systematically disadvantaging certain groups. For instance, a higher FPR for one group would mean that more individuals in that group are wrongly classified as high risk, potentially leading to harsher treatment.

Connection to Calibration

FPR and FNR parity help address an important limitation of calibration. While calibration ensures that risk scores are equally interpretable across groups, it does not control how errors are distributed.

As discussed earlier, a model can be perfectly calibrated (e.g., a 70% score means a 70% true risk for all groups) but still produce unequal error rates due to differences in base rates across groups. This can result in one group experiencing more false positives and another experiencing more false negatives.

By enforcing FPR and FNR parity, the model ensures that these errors are distributed more evenly across groups, which helps mitigate unfair treatment in decision outcomes.

Conclusion

Therefore, while calibration focuses on the fairness of predicted probabilities, FPR and FNR parity focus on the fairness of errors. Together, they highlight different dimensions of fairness, and in practice, it is often difficult to satisfy all of them simultaneously.

2. Prove that when P(Y =1 | A=0) ≠ P(Y =1 | A=1), no non-trivial classifier satisfies all three simultaneously.

For any non-trivial classifier with different base rates, calibration and equalized odds cannot both hold.

Assume that the two groups have different base rates:

- $\pi_0 = p(Y = 1 | A = 0) = 0.3$
- $\pi_1 = p(Y = 1 | A = 1) = 0.6$

This means that Group 0 has 30% actual positives and 70% actual negatives, while Group 1 has 60% actual positives and 40% actual negatives.

Suppose classifier satisfied both **FPR parity** and **FNR parity** hold :

- $TPR_0 = TPR_1 = 0.8$
- $FPR_0 = FPR_1 = 0.2$
- $FNR_0 \ (1 - TPR_0) = FNR_1 = 0.2$

Now, for a binary classifier, calibration at the decision level means that among people predicted positively, the actual probability of Y=1 should be the same across groups. In other words, can $P(Y = 1 | \widehat{Y} = 1, A = 0) = P(Y = 1 | \widehat{Y} = 1, A = 1)$?

We now calculate this quantity for each group.

- Group 0 :
  $P(Y = 1 | \widehat{Y} = 1, A = 0) \rightarrow$
  $formula = \frac{TPR}{TPR + FPR} = \frac{0.3 \times 0.8}{0.3 \times 0.8 + 0.7 \times 0.2} \approx 0.632$

- Group 1 :
  $P(Y = 1 | \widehat{Y} = 1, A = 1) \rightarrow$
  $formula = \frac{TPR}{TPR + FPR} = \frac{0.6 \times 0.8}{0.6 \times 0.8 + 0.4 \times 0.2} \approx 0.857$

Result :

$$P(Y = 1 | \widehat{Y} = 1, A = 0) \approx 0.632 \neq P(Y = 1 | \widehat{Y} = 1, A = 1) \approx 0.857$$

This means the classifier is **not calibrated across groups**, even though it satisfies equal FPR and equal FNR. Therefore, when base rates differ across groups, a non-trivial classifier cannot simultaneously satisfy calibration, FPR parity, and FNR parity. This example directly illustrates

3. Apply the impossibility result to the COMPAS empirical findings from Lecture 01. Which criterion did COMPAS satisfy? Which did it violate?

The results from Lecture 01 provide a clear example of the impossibility theorem, which states that when base rates differ across groups, a model cannot satisfy calibration, FPR parity, and FNR parity at the same time.

Calibration of recidivism rate

First, looking at calibration, we examine individuals with the same COMPAS score. For a decile score of 7, the observed recidivism rate is about 0.609 for African-American individuals and 0.602 for Caucasian individuals. These values are very close, which suggests that the same score corresponds to a similar level of risk across groups. This provides evidence that COMPAS satisfies calibration (predictive parity).

```
%%R
library(dplyr)

result_score7 <- df %>%
  mutate(
    two_year_recid_num = as.integer(as.character(two_year_recid))
  ) %>%
  filter(
    decile_score == 7,
    race %in% c("African-American", "Caucasian")
  ) %>%
  group_by(race) %>%
  summarise(
    n = n(),
    recid_rate = mean(two_year_recid_num, na.rm = TRUE),
    .groups = "drop"
  )

print(result_score7)
```

```
# A tibble: 2 × 3
  race                 n recid_rate
  <fct>            <int>      <dbl>
1 African-American   343      0.609
2 Caucasian          113      0.602
```

FPR & FNR

However, when we look at error rates, we see clear disparities. The false positive rate (FPR) for African-American individuals is 0.367, while for Caucasian individuals it is only 0.104. This means African-American individuals are more likely to be incorrectly classified as high risk. On the other hand, the false negative rate (FNR) is 0.173 for African-American individuals and 0.536 for Caucasian individuals, indicating that Caucasian individuals are more likely to be incorrectly classified as low risk. These differences show that both FPR parity and FNR parity are violated.

```
%%R
cat("\n\n── FPR and FNR Disparity by Race ──────────────────────────────────────\n")

disparity <- race_metrics %>%
  dplyr::select(race, n, FPR, FNR) %>%
  mutate(
    delta_FPR = round(FPR - FPR[race == "Caucasian"], 3),
    delta_FNR = round(FNR - FNR[race == "Caucasian"], 3)
  )

print(disparity)
```

```
── FPR and FNR Disparity by Race ──────────────────────────────────────
# A tibble: 6 × 6
  race                 n   FPR   FNR delta_FPR delta_FNR
  <fct>            <int> <dbl> <dbl>     <dbl>     <dbl>
1 African-American  3175 0.367 0.173     0.263    -0.363
2 Caucasian         2103 0.104 0.536     0         0
```

Overall, this example shows the impossibility of the result in practice. COMPAS maintains calibration, but violates the error rates across groups. In other words, the model ensures that the risk scores are meaningful and consistent, but it does not treat classification errors equally across different groups.

4. Propose a governance rule: under what circumstances should an organization prioritize calibration over FPR parity? Justify using the burden-shifting framework.

We can use the burden-shifting framework to decide when an organization should prioritize calibration over FPR/FNR parity, especially **in high-stakes settings like healthcare.**

Step 1: Plaintiff — Show the problem (statistical disparity)

First, the plaintiff points out that the model treats groups differently in terms of errors.
For example, consider a model that predicts the risk of adverse drug reactions:

Group A:
FPR = 25% (some patients are wrongly labeled as high-risk → unnecessary treatment)
FNR = 10%

Group B:
FPR = 10%
FNR = 25% (some patients are wrongly labeled as low-risk → missed treatment)

In simple terms, one group is more likely to be over-treated, while the other is more likely to be under-treated. This creates a clear fairness concern and establishes a prima facie case of disparity.

Step 2: Defendant — Explain why this might be acceptable (business necessity)

Now the responsibility shifts to the organization to explain why these differences might still be justified.

In medical settings, not all mistakes are equally serious.

- A false negative (FNR) means missing a real risk → a patient might not get needed treatment, which can be dangerous.
- A false positive (FPR) means overestimating risk → a patient might receive extra monitoring or mild treatment, which is usually less harmful.

Because of this, models are often designed to balance these risks and minimize overall harm, rather than force error rates to be exactly the same across groups.

Also, if different groups have different underlying risk levels (different base rates), then even a well-performing model will naturally produce different FPR and FNR. So unequal error rates do not automatically mean the model is "wrong", they may reflect real-world differences and clinical priorities.

Step 3: Plaintiff — Propose a better alternative (less discriminatory option)

Finally, the plaintiff must show that there is a better approach that reduces unfairness without sacrificing the model's purpose. In this case, a strong alternative is to prioritize calibration.

Calibration means that a risk score has the same real-world meaning for everyone. For example, if a patient is given a 70% risk score for a drug reaction, then about 70% of patients with that score should actually experience the outcome, regardless of their group.

Compared to FPR and FNR parity, calibration is more appropriate in medical contexts because decisions are based directly on predicted probabilities rather than classification labels. While FPR and FNR focus on balancing error rates, they do not guarantee that the risk scores themselves are accurate or interpretable. In contrast, calibration ensures that a given score (e.g., 70%) corresponds to the same real-world risk across all groups, which is essential for consistent and safe clinical decision-making. Therefore, when predictions are used as probability estimates rather than binary decisions, calibration provides a more meaningful and actionable notion of fairness.

This is especially important in healthcare, because doctors directly use these probabilities to make decisions (e.g., whether to prescribe a drug, adjust dosage, or monitor more closely).

Although enforcing equal FPR and FNR might reduce differences in error rates, it would typically break calibration due to differences in base rates (as shown by the impossibility result). This would make the risk scores unreliable, which is more problematic in medical contexts.

Therefore, in high-stakes settings where decisions rely directly on predicted probabilities, regulators should allow deviations from FPR/FNR parity if calibration is necessary for reliable decision-making and no less discriminatory alternative exists.